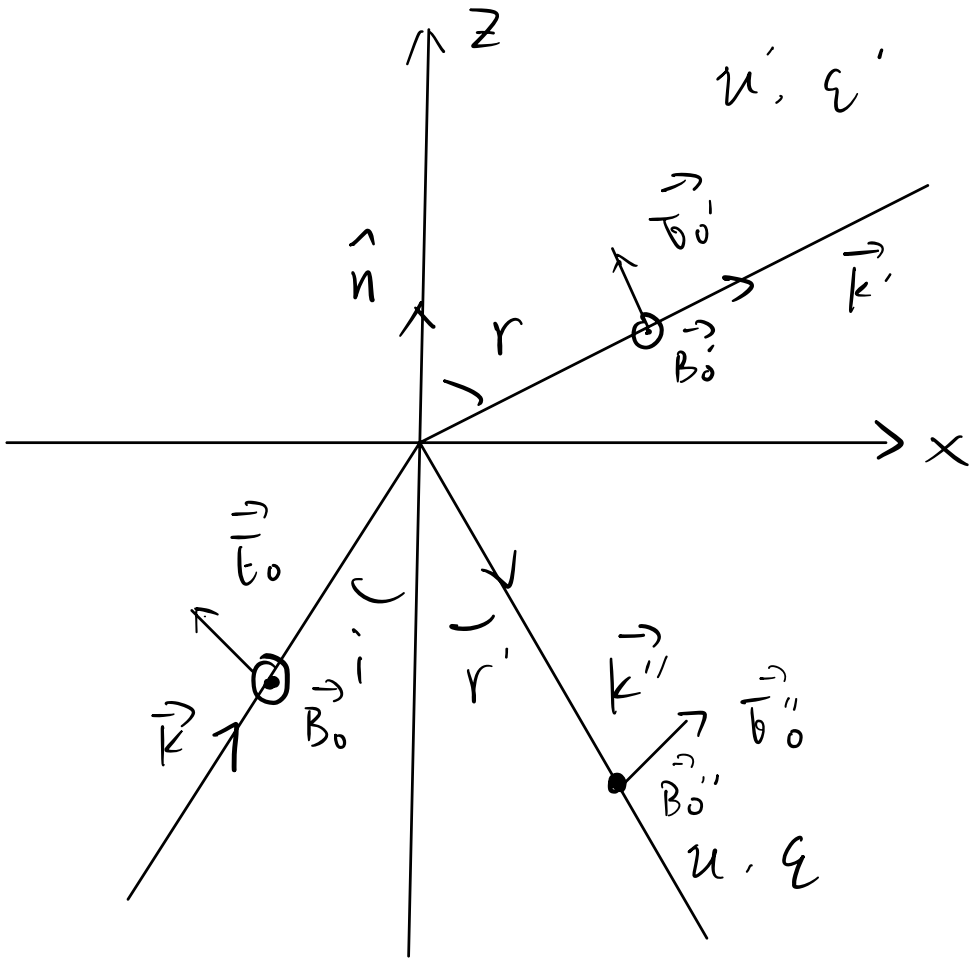
Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 04: Reflection and Refraction 2

Instructor: Zihan Wang

E in the plane of k and n (p polarized)



$$\begin{cases} [\varepsilon(\vec{E}_0+\vec{E}_0'')-\varepsilon'\vec{E}_0']\cdot\vec{n}=0 \\ (\vec{E}_0+\vec{E}_0''-\vec{E}_0')\times\vec{n}=0 \quad \checkmark \\ [\vec{k}\times\vec{E}_0+\vec{k}''\times\vec{E}_0''-\vec{k}'\times\vec{E}_0']\cdot\vec{n}=0 \\ \left[\frac{1}{u}(\vec{k}\times\vec{E}_0+\vec{k}''\times\vec{E}_0'')-\frac{1}{u'}(\vec{k}'\times\vec{E}_0')\right]\times\vec{n}=0 \quad \checkmark \end{cases}$$

④ $\Rightarrow \vec{H}_{//}+\vec{H}_{//}''=\vec{H}_{//}'$

$\Rightarrow \dfrac{E_0}{Z}+\dfrac{E_0''}{Z}=\dfrac{E_0'}{Z'}$

② $\Rightarrow \vec{E}_{//}+\vec{E}_{//}''=\vec{E}_{//}'$

$\Rightarrow E_0\cos i - E_0''\cos i = E_0'\cos r$

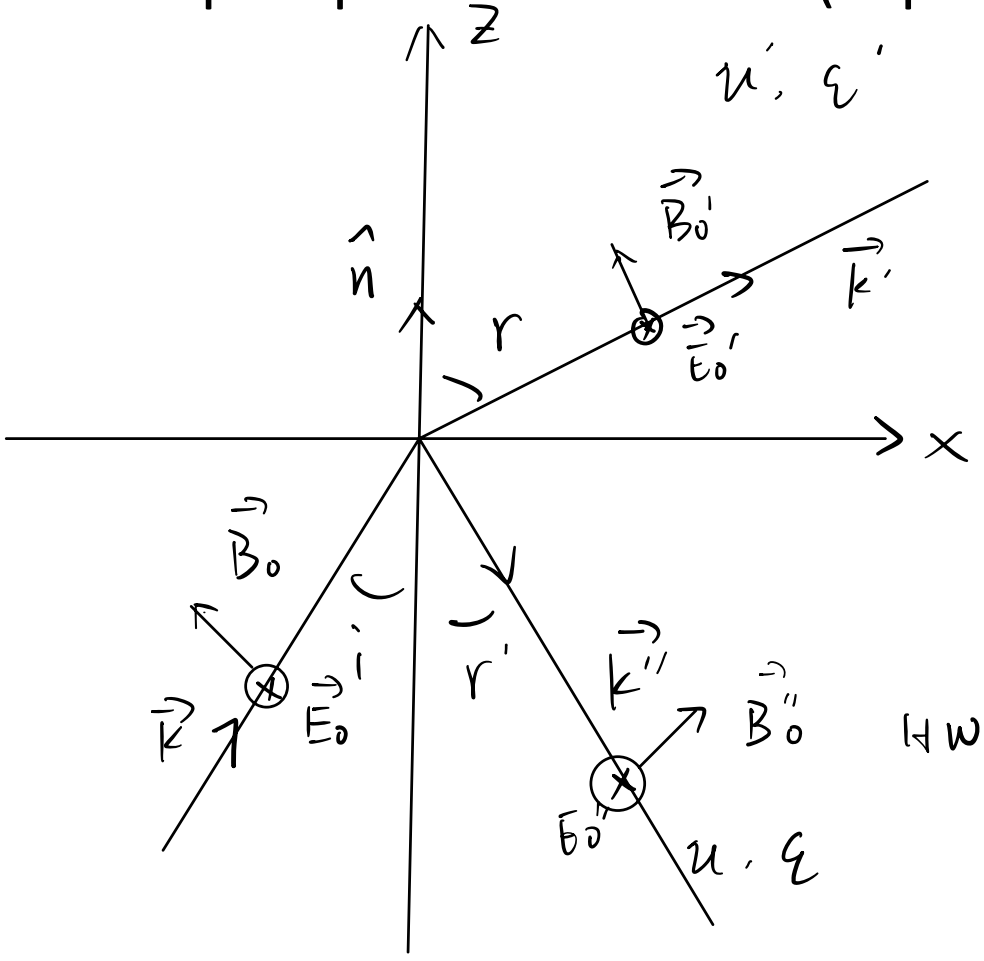
E in the plane of k and n (p polarized)

$$\begin{cases} \frac{E_0}{z} + \frac{E_0''}{z} = \frac{E_0'}{z'} \\ E_0 \cos i - E_0'' \cos i = E_0' \cos r \end{cases}$$

$$\Rightarrow \begin{cases} E_0'/E_0 = \dfrac{2z'\cos i}{z\cos i + z'\cos r} \\ E_0''/E_0 = \dfrac{z\cos i - z'\cos r}{z\cos i + z'\cos r} \end{cases}$$

Fresnel Equations

E perp to k and n (s polarized)



HW2

$$\begin{cases} [\varepsilon(\vec{E}_0 + \vec{E}_0'') - \varepsilon' \vec{E}_0'] \cdot \vec{n} = 0 \\ (\vec{E}_0 + \vec{E}_0'' - \vec{E}_0') \times \vec{n} = 0 \\ [\vec{k} \times \vec{E}_0 + \vec{k}'' \times \vec{E}_0'' - \vec{k}' \times \vec{E}_0'] \cdot \vec{n} = 0 \\ \left[\frac{1}{\mu}(\vec{k} \times \vec{E}_0 + \vec{k}'' \times \vec{E}_0'') - \frac{1}{\mu'}(\vec{k}' \times \vec{E}_0')\right] \times \vec{n} = 0 \end{cases}$$

$$\begin{cases} \dfrac{E_0'}{E_0} = \dfrac{2Z'\cos i}{Z'\cos i + Z\cos r} \\ \dfrac{E_0''}{E_0} = \dfrac{Z'\cos i - Z\cos r}{Z'\cos i + Z\cos r} \end{cases}$$

Energy transport

Reflection coef $R = \dfrac{|\langle \vec{S_R} \rangle \cdot \hat{n}|}{|\langle \vec{S_I} \rangle \cdot \hat{n}|}$

Transmission coef $T = \dfrac{|\langle \vec{S_T} \rangle \cdot \hat{n}|}{|\langle \vec{S_I} \rangle \cdot \hat{n}|}$

$\vec{S} = \vec{E} \times \vec{H} \qquad |\vec{E}| = Z|\vec{H}|$

$\Rightarrow \langle \vec{S} \rangle = \frac{1}{2} \frac{|\vec{E}|^2}{Z} \hat{k} \qquad \Rightarrow R = \left| \frac{E_0''}{E_0} \right|^2$

Energy transport

$$T = 1 - R$$

conservation of energy

$$T = \frac{\left| \langle \vec{S}_T \rangle \cdot \hat{n} \right|}{\left| \langle \vec{S}_I \rangle \cdot \hat{n} \right|} = \left| \frac{\tilde{E}_0'}{\tilde{E}_0} \right|^2 \frac{Z}{Z'} \frac{\cos r}{\cos i}$$

Example: Total Internal Reflection

- When n>n', r>i.
- What happens if sin r >1?

Snell's Law: $n \sin i = n' \sin r$

$$\sin r = \frac{n}{n'} \sin i$$

if $n > n'$, $\sin r$ can be larger than 1.

$$\cos r = \sqrt{1 - \sin^2 r}$$ if $\sin r > 1$.

$\cos r$ is imaginary.

$$i_0 = \arcsin \frac{n'}{n}$$ if $i > i_0$, TIR occurs.

Evanescent Plane Waves

$$\cos r = i\sqrt{\left(\frac{\sin i}{\sin i_0}\right)^2 - 1}$$

$$e^{i\vec{k}\cdot\vec{x}} = e^{ik(\sin r\, x + \cos r\, z)}$$

$$= e^{ik\sin r\, x}\, e^{ik\cos r\, z}$$

$$= e^{ik\sin r\, x}\, e^{-kz\sqrt{\left(\frac{\sin i}{\sin i_0}\right)^2 - 1}}$$

Propagating evanescent waves.

δ skin depth. $e^{-\frac{z}{\delta}}$ $\delta = \dfrac{1}{k\sqrt{\left(\frac{\sin i}{\sin i_0}\right)^2 - 1}}$

Phase shift by total internal reflection

S-pol $$\frac{\tilde{t}_0''}{\tilde{t}_0} = \frac{z'\cos i - z\cos r}{z'\cos i + z\cos r}$$

if $\cos r$ is imaginary,

$\Rightarrow$ $\tilde{t}_0''$ is complex.

$\Rightarrow$ $|\tilde{t}_0''| = \tilde{t}_0$ $\qquad \frac{\tilde{t}_0''}{\tilde{t}_0} = e^{-i\varphi}$

φ is the phase shift.

Poynting vector across the plane

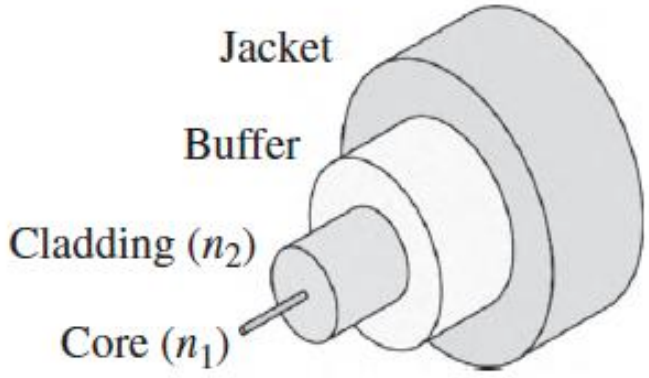
No energy flow can cross the plane although there are E and B beyond z=0.

$$\langle \vec{S}_T \rangle \cdot \vec{n} \propto \vec{k}' \cdot \vec{n} = k' \cos r \quad \text{is imaginary.}$$

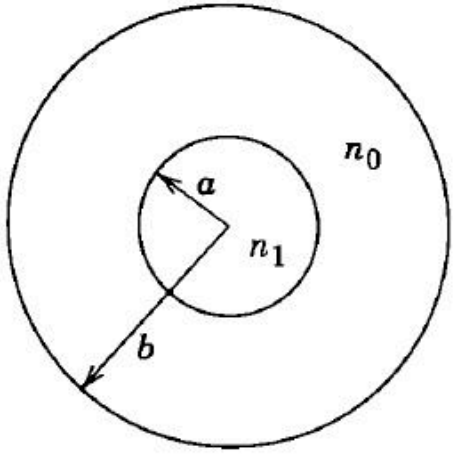
$$T = \left| \frac{\mathrm{Re}(\langle \vec{S}_T \rangle \cdot \vec{n})}{\mathrm{Re}(\langle \vec{S}_I \rangle \cdot \vec{n})} \right|$$

Application of total internal reflection: Optical Fiber

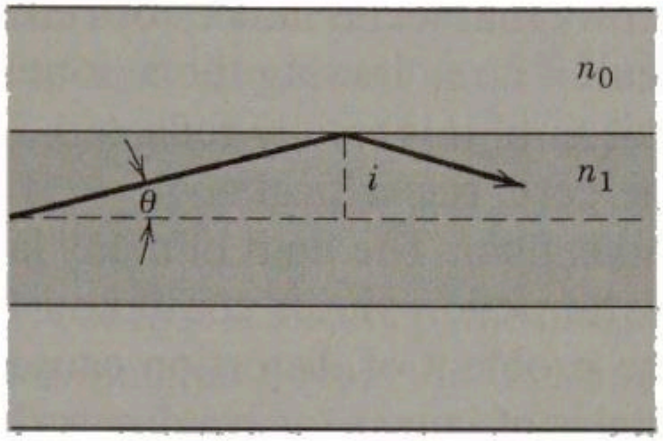
- Waveguides made entirely from dielectric materials are very desirable for applications where ohmic losses cannot be tolerated.
- The dominant high-speed, high-capacity telecommunications transmission medium.



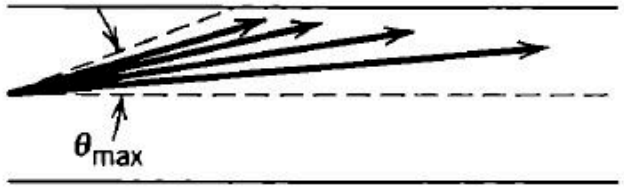
Optical Fiber



(a)



(b)



n0<n1, but the difference is very small, ~1%.

$$\theta_{max} = \cos^{-1}(n_0/n_1)$$

Example: Brewster's angle

- Assume $\mu = \mu'$
- For p polarized wave

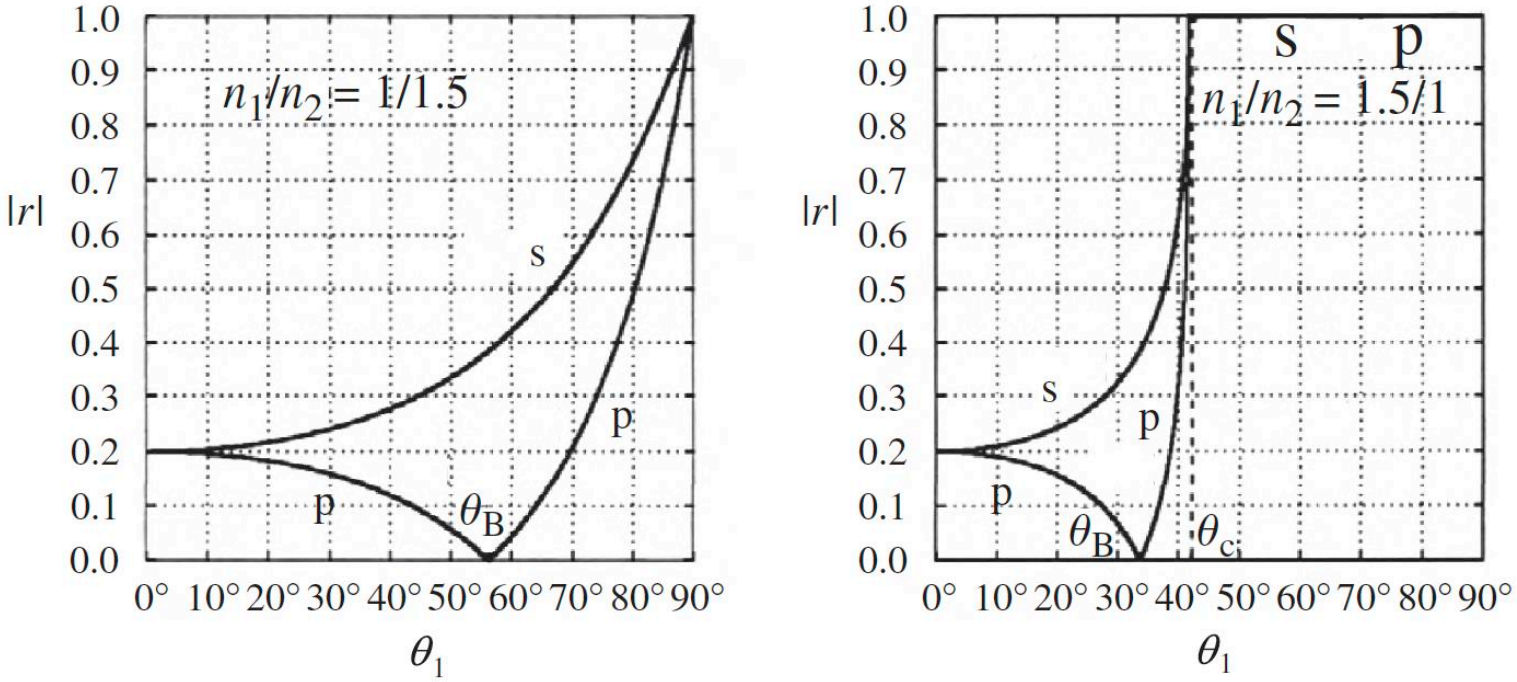
$$\frac{E_0''}{E_0} = \frac{Z\cos i - Z'\cos r}{Z\cos i + Z'\cos r}$$

$$\frac{E_0''}{E_0} = \frac{Z'\cos i - Z\cos r}{Z'\cos i + Z\cos r}$$

When $Z\cos i = Z'\cos r$, $E_0'' = 0$

HW2: $i_B = \arctan\frac{n'}{n}$

Brewster's angle

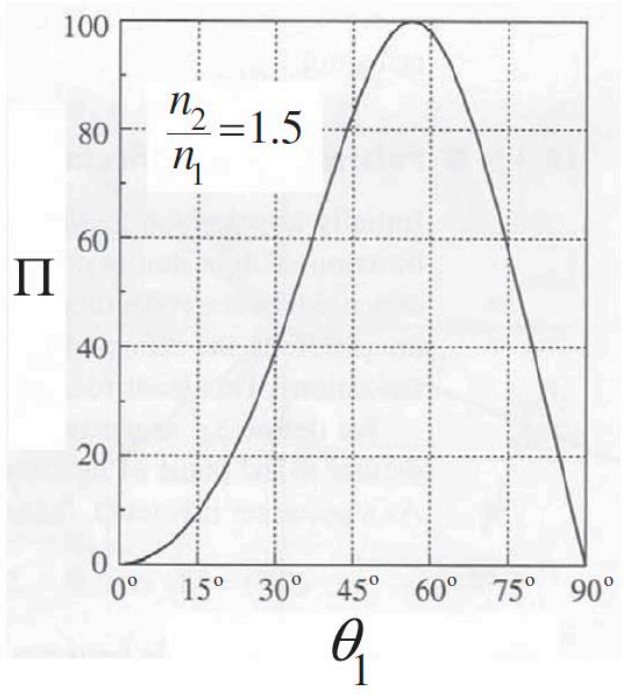


Relative polarization

$$\Pi = \left| \frac{R_s - R_p}{R_s + R_p} \right|$$

Π is 1. linear S pol

Π is 0. unpolarized.





Photographs taken of a window with a camera polarizer filter rotated to two different angles. In the picture at left, the polarizer is aligned to pass only the s polarization which is strongly reflected from the window. In the picture at right, the polarizer has been rotated 90° to eliminate the heavily polarized reflected sunlight, passing only the p polarization.